

VINAY YALAMANCHI

[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | yalamanchivinay9@gmail.com | +1(978)-726-0619 | Fremont, California, USA

CAREER OBJECTIVE:

AI Engineer and Computer Science Master's graduate with expertise in designing, developing, and deploying AI-powered software applications, scalable backend systems, and RESTful APIs using Python, FastAPI, PostgreSQL, SQL, C#, and Microsoft Azure. Experienced in building end-to-end AI solutions by integrating Machine Learning, Generative AI, Large Language Models (LLMs), and Natural Language Processing (NLP) into production-ready applications. Skilled in software engineering, cloud technologies, DevOps, CI/CD, Agile methodologies, and problem-solving, with a passion for developing intelligent, scalable, and high-performance software solutions.

SUMMARY OF SKILLS

- **Programming Languages:** Python, SQL, JavaScript, C#
- **Software Engineering:** Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful APIs, Software Development Life Cycle (SDLC), Software Testing, Unit Testing, Debugging, Code Reviews
- **Frameworks & Technologies:** FastAPI, React, Streamlit, HTML5, CSS3, JSON
- **Databases:** PostgreSQL, MySQL, SQLAlchemy
- **Cloud & DevOps:** Microsoft Azure, Azure DevOps, Docker, Git, GitHub, CI/CD Pipelines
- **Machine Learning & AI:** Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, Generative AI, Large Language Models (LLMs), Prompt Engineering, Natural Language Processing (NLP), Feature Engineering, Model Evaluation, Model Deployment
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Data Preprocessing, Exploratory Data Analysis (EDA)
- **Methodologies:** Agile Development, Scrum, Design Thinking, Hypothesis-Driven Development
- **Tools:** Visual Studio Code, Postman, Microsoft Excel, Microsoft Word, Microsoft PowerPoint

PROFESSIONAL EXPERIENCE

UL Solutions | Software Development Intern Fremont, California

Mar 2026 – Present

- Develop AI-enabled applications and backend services using Python, FastAPI, SQL, and cloud technologies.
- Design and maintain APIs, automation workflows, and data-processing solutions supporting engineering operations.
- Collaborate with cross-functional teams in Agile environments to deliver scalable software solutions.
- Implement CI/CD pipelines, testing, debugging, and deployment processes to improve software reliability.
- Integrate machine learning and data-driven capabilities into enterprise applications.
- Contribute to software design, code reviews, and technical documentation following engineering best practices.

Claro Software Solutions Pvt Ltd | Software Engineering Intern

Feb 2021 – Aug 2021

- Developed responsive web applications using HTML, CSS, and JavaScript while ensuring cross-browser compatibility and mobile responsiveness.
 - Implemented user authentication, order tracking, and payment-processing functionalities to enhance application capabilities and user experience.
 - Collaborated with development teams to gather requirements, implement features, and resolve technical issues in Agile environments.
 - Participated in software debugging, testing, and performance optimization activities to improve application reliability and functionality.
 - Assisted in sprint planning, stand-ups, peer reviews, and feature validation as part of the software development lifecycle.
 - Applied Design Thinking and Hypothesis Driven Development methodologies to improve usability and overall customer satisfaction.
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PROJECT'S

1. AI-Assisted Interview Simulator | Python, FastAPI, Streamlit, PostgreSQL, Hugging Face

- Designed and developed an AI-powered interview platform for role-specific mock interviews and automated feedback.
- Built RESTful APIs using FastAPI and integrated Speech-to-Text (STT), NLP, and semantic similarity analysis to evaluate candidate responses.
- Developed AI-driven analytics to assess response relevance, filler words, pauses, and confidence, providing personalized performance insights.
- Built interactive candidate and admin dashboards using Streamlit and PostgreSQL for interview management and reporting.
- Containerized the application with Docker and managed development using Git and GitHub.

2. Healthcare Prediction System | Python | Machine Learning | FastAPI | PostgreSQL

- Developed machine learning models to predict disease risk using healthcare datasets.
- Performed data preprocessing, feature engineering, model evaluation, and performance optimization.
- Built REST APIs for prediction services and integrated the models into a web application.
- Designed PostgreSQL databases for secure storage of patient and prediction data.

3. AI Recruitment Analytics Dashboard

- Developed analytics dashboards to monitor recruitment metrics, hiring trends, and candidate performance.
- Designed FastAPI backend services and PostgreSQL database models.
- Built interactive visualizations to support recruiter decision-making.

4. Block chain for the Trustworthy Decentralized Web Hosting

- Built a decentralized web hosting platform using block chain technology.
- Ensured data integrity and transparency across all users through block chain protocols.

5. End-to-End Conversion Speed Analysis of an FPT.AI-based Text-to-Speech Application

- Conducted performance analysis of text-to-speech conversion times using the FPT.AI API.
- Optimized data flow between local and remote servers to reduce processing time.

6. Internet of Things Projects: Smart Agriculture & Home Automation

EDUCATION

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| ➤ Master of Science in Computer Science, Rivier University, USA | - | Jan 2023 –2025 |
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CERTIFICATIONS

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| ➤ Data Analysis with R Programming | - | Google (May 2023) |
| ➤ Python for Data Science, AI & Development | - | IBM (Jun 2023) |
| ➤ Data Structures and Algorithms | - | Udemy (March 2025) |
| ➤ Claude Platform 101 | - | Anthropic (2026) |
| ➤ Claude Code in Action | - | Anthropic (2026) |
| ➤ Claude with the Anthropic API | - | Anthropic (2026) |